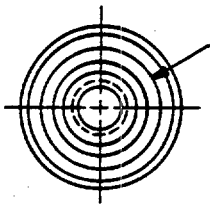
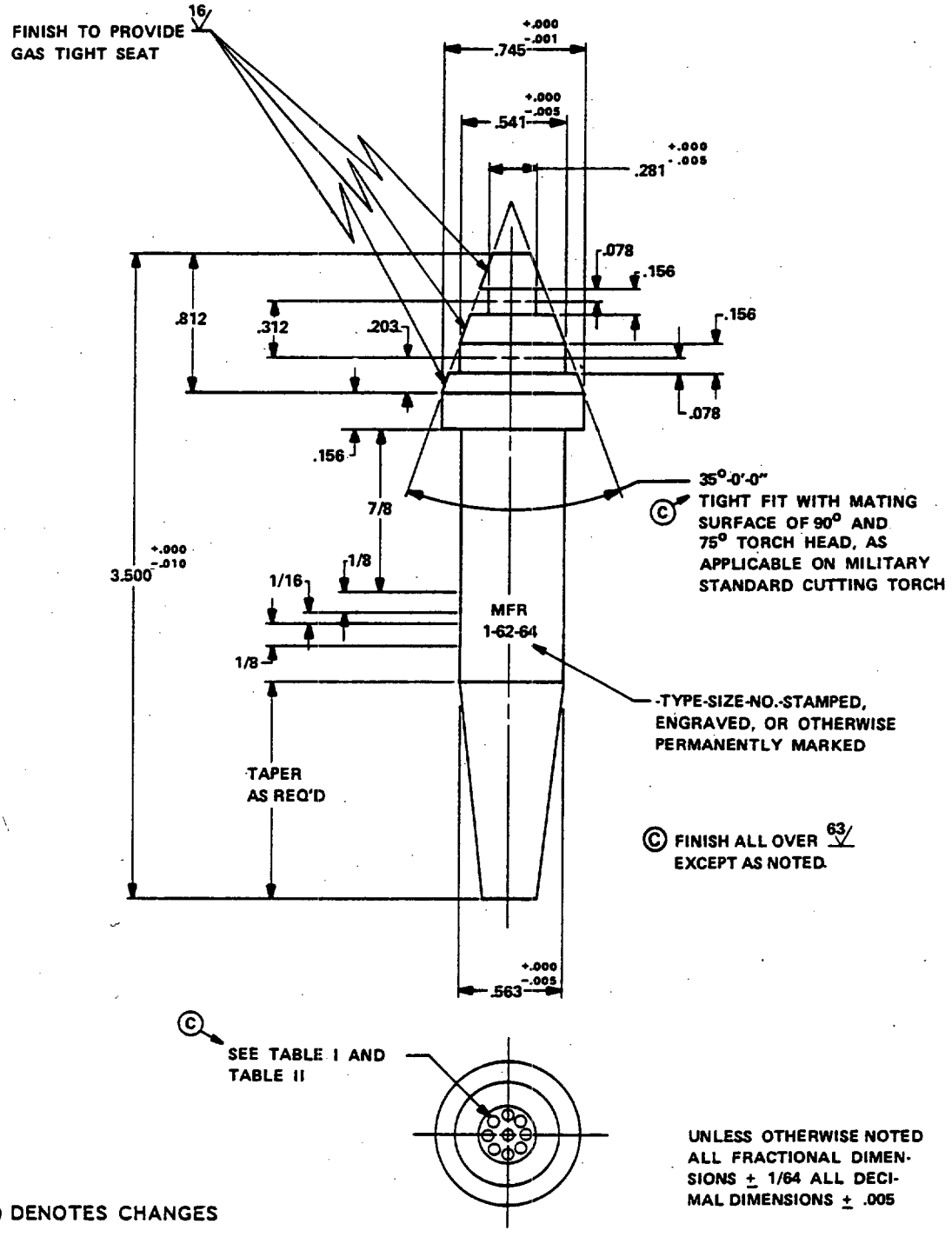
FED. SUP CLASS
3433

DO NOT SCALE DRAWING





UNLESS OTHERWISE NOTED
ALL FRACTIONAL DIMEN-
SIONS ± 1/64 ALL DECI-
MAL DIMENSIONS ± .005

© DENOTES CHANGES

APPROVED 28 SEPT 1951 REVISED (A) 26 NOV 1952 (B) 23 FEB 1973 © 18 AUG 1982

P.A YD Other Cust AR © 99	International interest	TITLE TIP FOR MILITARY STANDARD CUTTING TORCH	MILITARY STANDARD MS15013
Procurement Specification SEE NOTES 1 & 2 PAGE 5©		SUPERSEDES: MIL-T-16391 ©	PAGE 1 OF 5

DD FORM 1 MAY 73 672 (Coordinated) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

3433-0090

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© Materials:

1. Tip material shall be either 99.8 percent copper minimum, or tellurium copper (99.5 percent copper and 0.5 percent tellurium). Other materials that will perform equal or better may be used when authorized by the contracting officer.

Classification:

Type I - Tips for use with oxygen and acteylene.

Type II - Tips for use with oxygen and propane, or other fuel gases.

© Dimensions:

1. In addition to those dimensions shown in the drawing, tip dimensions shall conform to tables I and II, as applicable. Preheat and cutting orifices shall have a straight internal section not less than three-fourths of an inch. The removal of one-half inch of the tip end shall not change the performance or operational characteristics of the tip.

© General requirements:

1. Definition of terms as used in this section:

(a) Flashback. A recession of the flame into or back of the mixing chamber of the torch; in the case of tip mixing torches, flashback is usually characterized by the flame burning inside the tip with a hissing sound.

(b) Back-fire. The momentary recession of the flame into the torch tip followed by either immediate reappearance or complete extinguishment of the flame.

(c) Drag. The horizontal distance between the point of entrance and the point of exit of a cutting oxygen stream. The path of which is denoted by curved lines on the cut surface.

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© 2. Tip seats shall be free from nicks, scratches, and other surface irregularaties which would interfere with the seating of the tip. They shall be designed for cutting steel plate from 1/8 inch through 12 inches in thickness in accordance with tables I and II.

© 3. The flow rates shall be within 10 percent of the values shown in tables I, II, III, and IV as applicable. Cutting oxygen flow rate shall be made under the following conditions:

(a) Inlet pressures shall be measured at the inlet fitting of the torch by a gage accurate to ± 1 percent.

(b) Preheat valves shall be closed and no obstructions shall be permitted in the path of the emerging oxygen stream.

(c) Flow rates shall be measured with calibrated flowmeters or rotameters or other standard flow measuring devices.

© Table I - Sizes and dimensions of cutting tips, type I.

Tip Number	Preheat orifices		Cutting oxygen drill	Diameter preheat circle	Diameter external tip end	Cutting oxygen flow, cubic feet per hour (C.F.H.) at inlet pressure		Steel thickness range 1/
	Number	Drill size	Size	Inch	Inch	40 p.s.i.	60 p.s.i.	Inches
1-62-64	4	64	62	0.130	0.250	50	70	1/8 to 1/2
2-56-62	4	62	56	0.150	0.310	70	100	1/4 to 1-1/4
3-52-59	4	59	52	0.175	0.320	130	200	1 to 2-1/2
4-43-57	6	57	43	0.220	0.410	260	360	2 to 6
5-30-56	6	56	30	0.300	0.500	480	2/800	6 to 12

1/ For straight edge cutting of medium steel plate.
2/ At 80 pounds per square inch (p.s. i.) torch inlet pressure.

© Table II. Sizes and dimensions of cutting tips, type II.

Tip number	Preheat orifices		Cutting oxygen drill	Cutting oxygen flow C.F.H. at inlet pressure		Steel thickness range 1/
	Number	Shape	Size	40 p.s.i.	60 p.s.i	Inches
P-1-62	12	Slots	62	50	70	1/8 to 1/2
P-1-56	12	Slots	56	70	100	1/4 to 1-1/4
P-1-52	16	Slots	52	130	200	1 to 2-1/2
P-1-43	16	Slots	43	260	360	2 to 6
P-1-30	16	Slots	30	480	2/800	6 to 12

1/ For straight edge cutting medium steel plate.
2/ At 80 p.s.i. torch inlet pressure.

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© 4. When attached to a military standard torch, tips shall:

(a) Produce smooth, flat, high quality cuts in steel, with drag of not more than ten percent of plate thickness when making cuts under conditions indicated in tables III and IV.

(b) When starting from a neutral flame and the cutting oxygen valve closed, not cause flashback when first the oxygen and then the gas is reduced alternately in ten percent increments until insufficient gas is available to sustain flame. The procedure shall be repeated with the gas instead of oxygen being reduced first.

(c) Produce a flame, with the cutting oxygen valve closed and a neutral preheat flame, that will continuously pierce a minimum of 1/16-inch thick plate, in the first six inches of travel, at travel speeds indicated in tables III and IV. The flame shall be a maximum of 3/32-inch from the plate during this operation. Preheat flames shall be not less than 3/8-inch long and pointed.

(d) Indicate no leakage around the tip nut when the torch is lighted and cutting oxygen flowing at the amount shown in table III and IV and the torch is submerged in water. A leak is indicated by the presence of bubbles coming from around the tip nut.

(e) When adjusted to a neutral preheat flame, not backfire when the tip end is brought in contact with a steel plate at an 85° angle and then slowly raised to a 90° angle.

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P.A YD Other Cust © AR 99	International interest	TITLE TIP FOR MILITARY STANDARD CUTTING TORCH	MILITARY STANDARD MS15013
Procurement Specification SEE NOTES 1 & 2 PAGE 5 ©		SUPERSEDES: MIL-T-16391 ©	PAGE 4 OF 5

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© Table III - Performance standards, type I.

Tip number	Cutting performances			Preheat performance	
	Oxygen inlet pressure p.s.i.	Thickness cut, inches	Speed of cut inches per minute (min.)	Preheat oxygen consumption, C.F.H.	Speed of travel inches per minute (min.)
1-62-64	40	1/2	12	10 to 15	9 to 10
2-56-62	55	1	8-1/2	10 to 15	10-1/2 to 11-1/2
3-52-59	60	2-1/2	5-1/2	16 to 20	14 to 16
4-43-57	60	6	4	29 to 33	21 to 23
5-30-56	65 to 70	12	2 to 3	60 to 70	26 to 28

© Table IV - Performance standards; type II.

Tip number	Cutting performances			Preheat performance	
	Oxygen inlet pressure p.s.i.	Thickness cut, inches	Speed of cut inches per minute (min.)	Preheat oxygen consumption, C.F.H.	Speed of travel inches per minute (min.)
P-1-62	40	1/2	12	30 to 40	14 to 16
P-1-56	55	1	8-1/2	35 to 45	15 to 17
P-1-52	60	2-1/2	5-1/2	45 to 55	17 to 19
P-1-43	60	6	4	60 to 70	21 to 23
P-1-30	65 to 70	12	2 to 3	80 to 100	27 to 31

© **Marking:**

1. Tips shall be clearly and permanently marked with the type and size, and manufacturer's name or trademark located as shown in the drawing. The type and size shall be as indicated in tables I and II in tip number column as applicable.

© **Preparation for delivery:**

1. Each tip shall be individually packaged in a cylindrical cartridge type cardboard container.

2. Tips shall be packed and marked for shipment conforming to commercial standard practices.

© Part number:

The part number shall consist of this document number plus the number of the tip desired from the tip-number column of Table I or II.

Example: MS15013-1-62-64 for tip number 1 of Table I

© Notes: 1. Subject tip is intended to fit the standard military torch which was covered by MIL-T-16183 (MIL-T-16183 was cancelled 14 Feb. 1980). Many of these torches remain in use throughout DoD, therefore the tip will be required for replacement until all torches of this type are removed from use.

2. MIL-T-16391, which has been cancelled, was the procurement document for MS15013. Performance and design requirements of MIL-T-16391 have been incorporated in this revision.

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Other Cust AR 99 ©		TIP FOR MILITARY STANDARD CUTTING TORCH	MS15013
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